

2. Graphically Summarizing Quantitative Data

Consider the data on the right

- a** Find the number of classes needed to construct a histogram.
- b** Find the class length.
- c** Define nonoverlapping classes for a frequency distribution.
- d** Tally the number of values in each class and develop a frequency distribution.
- e** Draw a frequency histogram for these data (hand-drawing is fine).
- f** Develop a percent frequency distribution (hand-drawing is fine).

36	46	40	38
39	40	38	38
36	42	33	34
35	34	37	37
36	41	22	17
20	36	33	25
19	42	28	38

The largest is 46

The smallest is 17

Step 1: Number of classes

- In **example 2.2**, $n = 28$
 - For $K = 4$, $2^4 = 16, < n$
 - For $K = 5$, $2^5 = 32, > n$
 - So use $K = 5$ classes

Step 2: Class length

- $(46-17) / 5 = 5.8$
 - Because number in integers, round **up** to 6

Step 3: Form non-overlapping classes of equal width

Class 1	17 and less than 23 (i.e., 17, 18, 19, 20, 21, 22)
Class 2	23 and less than 29 (i.e., 23, 24, 25, 26, 27, 28)
Class 3	29 and less than 35 (i.e., 29, 30, 31, 32, 33, 34)
Class 4	35 and less than 41 (i.e., 35, 36, 37, 38, 39, 40)
Class 5	41 and less than 47 (i.e., 41, 42, 43, 44, 45, 46)

Step 4: Tally and count the number of measurements in each class

Class 1	4
Class 2	2
Class 3	4
Class 4	14
Class 5	4

Step 5: Graph the histograms

